Generation Structure and Coupling Matrix from GF(27)

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Abstract

The four irreducible polynomials $f_1, \dots, f_4$ over GF(3) define the physical orbits in GF$(27)^*$. Their root orbits are genuine Frobenius orbits under $k \mapsto 3k$ mod 26; the three elements correspond to the three generations. The trace bilinear form $\langle a, b \rangle = \mathrm{Tr}_{\mathrm{GF}(27)/\mathrm{GF}(3)}(ab)$ gives the coupling structure: $f_3 \times f_4$ (charged leptons × quarks) is a permutation matrix **[B]**; $f_1 \times f_3$ (neutrinos × charged leptons) has normalised entries $1/\sqrt{2}$ **[B]**. Mixing angles (CKM, PMNS) themselves remain **[S]**.

.1 Why CKM mixing angles are complex

The CKM matrix V_{CKM} is unitary with four independent parameters (Wolfenstein parametrisation): $\lambda \approx 0.225$, $A \approx 0.811$, $\rho \approx 0.124$, $\eta \approx 0.356$. The CP-violating phase $\delta_{CP} \approx 68^\circ$ ($e^{i\delta_{CP}} = 0.375 + 0.927i$) makes the matrix essentially complex. It is responsible for CP violation in kaon and B-meson decays (observed 1964 and 2001 respectively).

What GF$(27)^*$ provides and what it does not

Gal(GF(27)/GF(3)) $= \mathbb{Z}_3$ has three irreducible representations over $\mathbb{C}$: trivial (1), $\omega = e^{2\pi i/3}$ (120°), ω^2 (240°). If the coupling matrix transforms under ρ_1, a complex phase arises — the most plausible Galois candidate for the CP sector.

The problem: ω corresponds to 120°, not 68° (gap 52°) — no algebraically forced candidate.

Complete structure of the open problem [S]

Three steps are needed for the full CKM matrix from GF(27): (1) **Real zero structure** — which couplings vanish: the trace bilinear form gives the permutation matrix (Theorem B) **[B]**. (2) **Mixing angles** — corrections to the permutation matrix giving $\lambda \approx 0.225$, $A \approx 0.811$: no forced candidate in GF(27) **[S]**. (3) **CP phase** — a complex embedding of the Galois group forcing $\delta_{CP} \approx 68^\circ$: $\mathbb{Z}_3$ phases give 120°, not 68° **[S]**.

Core problem: GF$(27)^*$ is a finite field over GF(3) with no natural complex structure. The trace bilinear form gives values in $\{0, 1, 2\} \subset$ GF(3). A complex embedding forcing $\delta_{CP} = 68^\circ$ does not currently exist **[S]**.

Bibliography

[1] J. Pascher, Doc. 347, FFGFT corpus (Sept. 2026).

[2] J. Pascher, Doc. 346, FFGFT corpus (Sept. 2026).